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IDeT-Sat: a Cubesat Design for Space Debris Detection and Analysis

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Abstract

Currently, there is a significant amount of space debris in orbit. Although many larger debris are monitored and recorded, most smaller debris under 10 cm are not tracked. Despite their size, these small debris still pose significant threats due to their high velocity. Therefore, detecting and monitoring small space debris is vital. To achieve this goal, Identification of Debris & Telemetry Satellite (IDeT-Sat) has been developed by the University of Nottingham student team for the UK Students for the Exploration and Development of Space (UKSEDS) and Space & Satellite Professionals International (SSPI) Satellite Design Competition 2022-23. It is a 3U CubeSat platform with the objective of monitoring space debris and promoting space sustainability by gathering data on space debris 1cm-10cm in size with stereovision cameras. To achieve this objective, a payload system consisting of three cameras has been designed for debris detection, data acquisition and debris property analysis. A CubeSat platform has also been developed to support the payload, and an engineering model for the platform has been constructed with commercial-off-the-shelf (COTS) parts. All systems are designed based on competition rules and requirements. Through testing, an asynchronous problem was observed between payload cameras, leading to the failure of the debris detection and data acquisition phase. The suspected cause for this problem is camera adapter software issue. Currently, the team is working on resolving the issue through examination and remediation of software, while also considering alternative adapter options. The experiment on debris analysis yielded promising results, indicating that the designed system is feasible. However, measurement error could be further reduced. Improvements that can be made in the future include resolving camera asynchronous issue, improving measurement accuracy, debris matching through feature matching techniques, developing camera calibration software, and identifying more debris features, including average diameter, colour, shape and albedo.
Keywords: Student Team Competition; CubeSat; Space Debris Detection and Analysis

Nomenclature

Symbol(s)	Variable(s)	Unit
X, Y	Cartesian coordinate of debris on image	N/D
f	Camera focus length	m
N_x, N_y	Total number of pixels in x, y direction	N/D
S_x, S_y	Camera sensor size in x,y direction	m
S_{px}	Pixel size	m

X_{f-i}, Y_{f-i}	Coordinate of debris image on sensor (i = camera number)	m
O	Center point between Camera 2 and 3	N/D
x, y, z	Debris distance from CubeSat in x, y, z directions	m
D	Debris distance to CubeSat	m
d	Distance between cameras	m
$(FOV)_x, (FOV)_y$	Camera field of view in x, y direction	Rad
$\lvert\underline{v}\rvert$	Magnitude of debris velocity	m/s
$\underline{\hat{v}}$	Direction of debris velocity	N/D
dx, dy, dz	Debris position differences between line blocks	m/s
dt	Time interval between debris crossing adjacent line blocks	m/s
$\underline{\hat{x}}, \underline{\hat{y}}, \underline{\hat{z}}$	Unit vector in x, y, z direction	m/s

Acronyms/Abbreviations

ADCS	Attitude Determination and Control System
BPCB	Battery Protection Circuit Board
COTS	Commercial-off-the-shelf
EPS	Electric Power System
IDeT-Sat	Identification of Debris & Telemetry Satellite
IMU	Inertial Measurement Unit
OBDH	On-board Data Handling system
PLA	Polylactic Acid
SIFT	Scale-Invariant Feature Transform
SSPI	Space & Satellite Professionals International
SURF	Speeded Up Robust Features
TT&C	Telecommunication, Tracking and Command system
UKSEDS	UK Students for the Exploration and Development of Space
USSSN	US Space Surveillance Network

1. Introduction

Ever since the launch of Sputnik in 1957, humans have been continuously sending objects into orbit. For a long period of time, space sustainability has been a neglected topic and many satellites were left in space after they were defuncted [1]. This leads to a significant amount of space debris in orbit, which can damage active spacecraft upon collision. To track these space debris and other artificial object in space, the US Department of Defence has established a cataloguing system named US Space Surveillance Network (USSSN) [2]. As of August 2024, USSSN regularly checks over 36,000 space objects, most of which are space debris, and keeps them in the catalogue [3]. These catalogued debris only include those that are big enough to be detected. Smaller debris with lengths shorter than 10 cm are not catalogued and monitored [2]. However, small size doesn't prevent debris from posing serious threats to operating spacecrafts. With their high speed, these small debris can still significantly damage spacecrafts in service [1]. The Aerospace Corporation illustrated this danger as: "An aluminium sphere 1 cm in diameter is comparable to a 400-lb safe travelling at 60 mph" [1]. Therefore, monitoring and tracking these small debris is crucial.

To tackle this issue, IDeT-Sat has been designed by the University of Nottingham student team for the UKSEDS and SSPI Satellite Design Competition 2022-23. It is a 3U CubeSat platform developed based on competition rules and requirements. It aims to monitor space debris and promote space sustainability by gathering data on small space debris 1 cm - 10 cm in size with stereovision cameras. The requirements for IDeT-Sat on target debris properties were determined based on [1] and are summarised in Table 1.

Table 1. Property of target debris

Distance*	Velocity	Size
1 - 2.5 m	0.01 – 1 m/s	1 – 10 cm

*Distance = Debris distance from CubeSat

This paper will present the preliminary design and test results of the IDeT-Sat, summarize the achievements and failures during the development, and discuss the improvements that can be made in the future.

2. System Design

In this section, the design of the payload and CubeSat platform will be summarized.

2.1 Payload design

IDeT-Sat payload utilizes three cameras for debris detection and data acquisition, as well as Raspberry Pi from the On-board Data Handling system (OBDH) for data analysis, as shown in Fig. 1. Here, Raspberry Pi Camera Module 3 is chosen for cameras because of its high resolution and adequate field of view and the adapter is chosen as Arducam Multicamera Adapter Module V2.2.

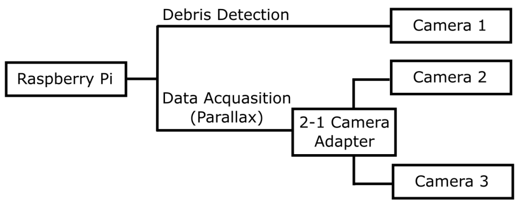

Fig. 1. Payload schematic

2.1.1 Debris detection

For debris detection, Camera 1 constantly records a video. The Raspberry Pi then analyses this video to identify the debris when it passes a manually set detection line block, as shown in Fig. 2. Debris is detected when pixels' brightness is outside of a set number of standard deviations from the mean brightness of the detection line at that time.

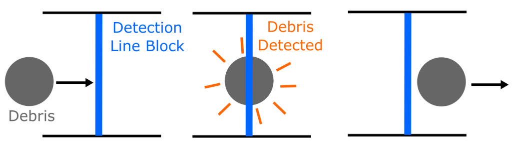

Fig. 2. Detection line block mechanism

Here, each detection line block consists of multiple lines, as shown in Fig. 3, to compensate for the fast speed of debris and limited frame rate of the camera. Considering the maximum frame rate of Raspberry Pi Camera Module 3 and the target debris properties list in Table 1, it is concluded that 40 FPS camera frame rate setting, along with 16 detection lines with 2-pixel separation between each detection line, is adequate to cover all possible combinations of debris size, distance, and velocity, to guarantee at least one detection per line block.

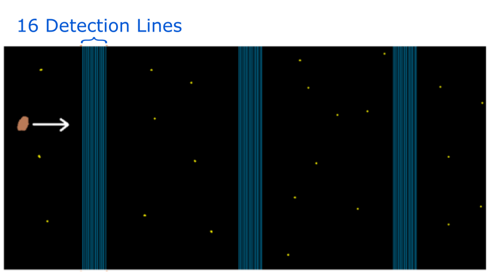

Fig. 3. Depiction of line blocks

2.1.2 Data acquisition

Upon every detection of debris, Cameras 2 and 3 will simultaneously take a photo each, forming a pair of photos. These pairs of photos will then be stored in the Raspberry Pi for data analysis. The setup for Camera 2 and 3 are shown in Fig. 4.

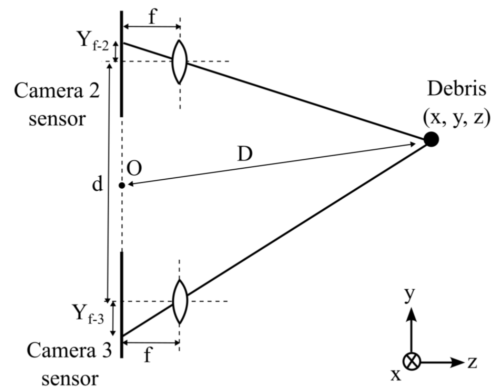

Fig. 4. Data acquisition cameras setup

Due to multiple detection lines within one line block, multiple photos might be taken for the same debris within one line block, especially for large debris with slow velocity. This might cause IDeT-Sat to identify one debris as multiple debris. To avoid this issue, the OBDH will compare the height coordinates of debris in the photos. This examines the similarity between photos and will allow OBDH to categorize and group the images for corresponding debris.

2.1.3 Data analysis

Data analysis includes determining the location, distance, size, and velocity of the debris. Here, all numerical values are averaged using all images of the same debris for higher accuracy.

Debris location upon every detection is determined via parallax. To begin with, the debris position within each photo, expressed in number of pixels from edge, is calculated using the OpenCV Template Matching technique. Here, the debris image is first cropped out from the detection image with the maximum allowed size. This cropped debris image is then converted to greyscale to generate a template image. After that, the template image is analysed against the photo in which the location of the debris needs to be determined. The convolution between the template image and the detection line along a fixed x position in the photo is calculated with the TM_CCOEFF_NORMED enumerator. The debris position within the photo is said to be at the convolution's global maximum. An illustration of this step is shown in Fig. 5.

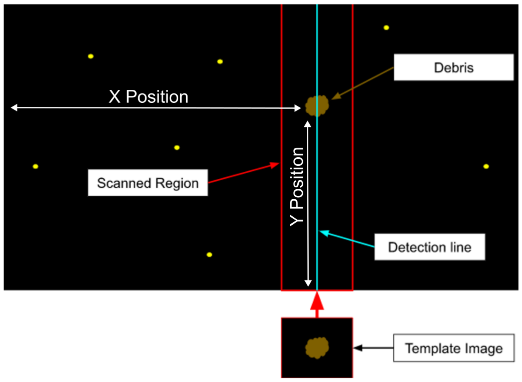

Fig. 5. Template Matching technique illustration

With the debris location identified in the photos taken, its coordinate, (X, Y), can be defined using a Cartesian coordinate system with origin at the image centre. Hence, following the method provided by [4], the physical location of the debris image projected on the camera sensor can be calculated as:

$$X_{f-i} = \frac{X \cdot S_x}{N_x} \quad [1]$$

$$Y_{f-i} = \frac{Y \cdot S_y}{N_y} \quad [2]$$

Adhering to the method outlined by [4], the distance of debris from the origin of the CubeSat in x, y, z direction can then be calculated as:

$$x = \frac{d}{2}\frac{X_{f-2} + X_{f-3}}{Y_{f-2} - Y_{f-3}} \quad [3]$$

$$y = \frac{d}{2}\frac{Y_{f-2} + Y_{f-3}}{Y_{f-2} - Y_{f-3}} \quad [4]$$

$$z = \frac{d \cdot f}{Y_{f-2} - Y_{f-3}} \quad [5]$$

Therefore, the distance between the debris and the centre point O between Camera 2 and 3 can be found as:

$$D = \sqrt{x^2 + y^2 + z^2} \quad [6]$$

Debris size is determined using the pixel size and the size of debris in pixels. Here, pixel size is the size that each pixel corresponds to in the real world. The overall pixel size is calculated using the average of pixel size in the x and y directions, as:

$$S_{px} = D \cdot \left(\frac{\tan\left(\frac{(FOV)_x}{2}\right)}{N_x} + \frac{\tan\left(\frac{(FOV)_y}{2}\right)}{N_y}\right) \quad [7]$$

The size of debris in the pixels is found through edge detection, which is applied to a greyscale version of the image cropped around the debris' likely position for accelerating the computation process. Two contour lines are returned, enclosing the edge of the debris at their midpoint. Here, Laplacian operator is used. This is because it applies 2^{nd} order spatial derivative across the image, resulting in narrower, more exaggerated detection peaks, and hence less false edge detection and better results, comparing to other operators such as Sobel edge detector or Canny edge detector [5–7].

With edge detection, the number of pixels contained within the edge contour line can be found. By multiplying it by pixel size corresponding to the debris distance, the size of debris can be calculated.

For debris velocity analysis, it is calculated using the time of flight between the blocks of lines. Calculated velocity is represented by a magnitude and a directional unit vector, as

$$|\underline{v}| = \sqrt{\frac{(dx)^2 + (dy)^2 + (dz)^2}{(dt)^2}} \quad [8]$$

$$\hat{\underline{v}} = \frac{dx\hat{\underline{x}} + dy\hat{\underline{y}} + dz\hat{\underline{z}}}{\sqrt{(dx)^2 + (dy)^2 + (dz)^2}} \quad [9]$$

2.2 CubeSat platform design

IDeT CubeSat is a 3U CubeSat platform. The structure is custom-designed with Aluminium 6061 railing for the primary structure and 3D-printed Polylactic Acid (PLA) plates and stand-offs for the secondary structure. The schematic diagram for CubeSat structural layout can be found in Fig. 6.

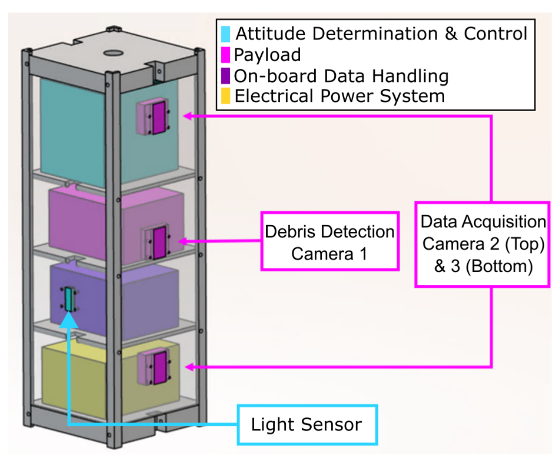

Fig. 6. Schematic diagram for CubeSat layout

The OBDH consists of 2 Raspberry Pi 3A's with a Master-Slave configuration. Its integration with other subsystems is shown in Fig. 7.

For Telecommunication, Tracking and Command system (TT&C), it is designed based on the Wi-Fi module inbuilt in the Raspberry Pi, following the requirement of UKSEDS and SSPI Satellite Design Competition [1].

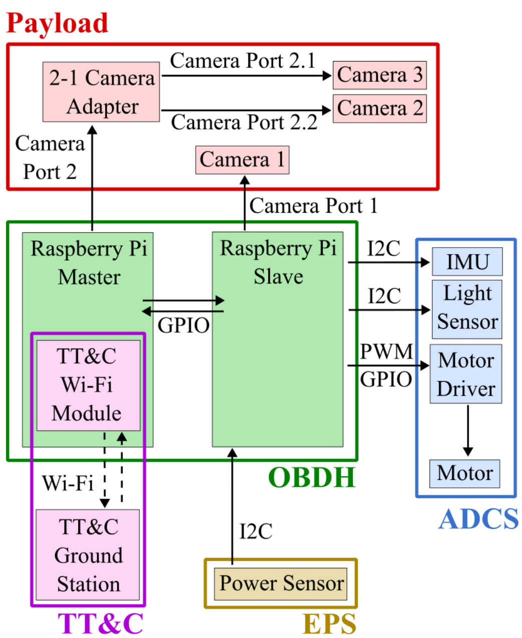

Fig. 7. System integration diagram

For Attitude Determination and Control System (ADCS), it utilizes light sensor and Inertial Measurement Unit (IMU) for attitude determination and employs reaction wheel and P-controller for attitude control. It is tailored for the UKSEDS and SSPI Satellite Design Competition where CubeSat is only required to rotate around the longitudinal axis [1].

The Electric Power System (EPS) utilizes Lithium-Ion batteries, each containing a dedicated Battery Protection Circuit Board (BPCB), for power supply.

3. Results

The results obtained and issues found during the verification of payload functions will be presented in this section.

3.1 Debris detection & Data acquisition

System is designed to use a multi-camera adapter to take photos simultaneously for distance calculation. The largest acceptable time interval between images taken is 0.05s. However, through testing, it was found that the time interval between the image taken by Camera 2 and Camera 3 could only be reduced to 1 second, which is larger than the maximum acceptable interval, causing issues for data analysis.

3.2 Debris analysis

Due to the unresolved issue discussed in section 3.1, and considering the timeframe of UKSEDS and SSPI Satellite Design Competition, an alternative debris detection and data acquisition setup was used for debris analysis experiments. This temporary experiment setup used only two identical cameras to shoot video, with no third camera for debris detection. Debris is detected from the videos via background subtraction, with sizes much above or below expectation discarded. The identified debris is then tracked by matching centroid positions of the bounding box of debris detected between frames.

Based on this temporary setup, an engineering model of IDeT-Sat has been constructed with COTS parts, as shown in Fig. 8. Using this engineering model, the designed methodology and software code for debris analysis were tested with sample debris through eight experiments. The results of these experiments are shown in Table 2.

Table 2. Experiment results for debris analysis

No.	Distance (m)		Velocity (m/s)		Height (cm)		Width (cm)	
	Actual	Measured	Actual	Measured	Actual	Measured	Actual	Measured
1	1.6	1.23	0.83	0.64	3	1.64	4	2.91
2	1.6	1.32	0.83	0.68	6	2.81	8	2.86
3	1.8	1.54	0.79	0.72	3	1.37	4	2.86
4	1.8	1.61	0.79	0.69	6	2.43	8	3.99
5	2.2	1.72	0.47	0.34	3	1.52	4	2.34
6	2.2	1.69	0.47	0.35	6	1.47	8	3.12
7	2.5	2.23	0.62	0.43	3	1.21	4	2.56
8	2.5	2.12	0.62	0.52	6	2.12	8	3.43

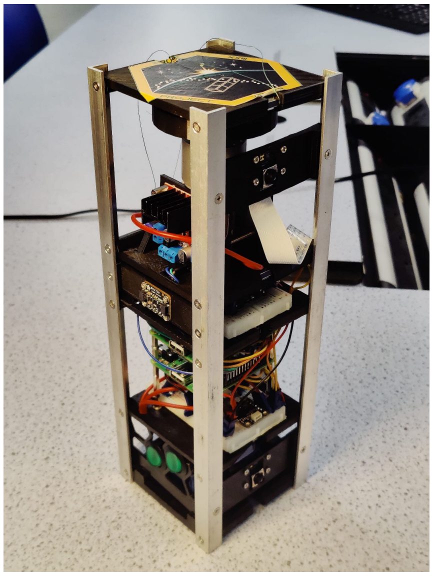

Fig. 8. Engineering model for debris analysis experiments

4. Discussion

The overly large time interval between images taken indicates that the current design is incomplete for debris detection and data acquisition tasks. The suspected reason for this problem is camera adapter software issue. Currently, the team is working on resolving the issue through examination and remediation of the software, while also considering alternative adapter options such as Pimoroni Camera Cable Adapter CAM007 or Waveshare Camera Scheduler 16740.

The experiments for debris analysis yielded results that were close to expected, indicating that the designed system is feasible and that debris properties can be identified using the designed method. However, significant errors are observed between actual values and measured values. This might be due to the calculated pixel size not being accurate enough, resulting in large errors in debris instantaneous position identification. Hence, further investigation and improvement on the pixel size calculation method is needed.

For other refinements on data acquisition, debris matching can be further improved by introducing feature matching techniques, such as Scale-Invariant Feature Transform (SIFT) algorithm or Speeded Up Robust Features (SURF) algorithm. Such improved debris matching can better compare the similarity between detected debris and hence facilitate debris categorization and grouping. Camera calibration software would also need to be developed. Such software should distort the images using a calibration matrix, and hence compensate for the fact that data acquisition cameras are not perfectly parallel in reality.

For other refinements on data analysis, more features of debris can be measured. Debris average diameter can be found by summing X and Y separations along the contour and averaging these separations. Debris colour can be found by averaging the RGB values of pixels contained within the contour. Debris shape can be categorised by edge positions from the contour. Potentially, the albedo of the debris could also be measured. This could be done through estimating the brightness of the sun at the debris, and comparing it with the brightness of light reflected from the debris measured by the camera.

5. Conclusions

IDeT-Sat is a 3U CubeSat platform designed the University of Nottingham student team for the UKSEDS and SSPI Satellite Design Competition. It aims to promote space sustainability by gathering data on small space debris 1 cm - 10 cm in size with stereovision cameras.

The development of IDeT-Sat consists of both software and hardware. A payload system consisting of three cameras has been designed for debris detection, data acquisition and debris property analysis. A CubeSat platform has also been developed to support the payload, and an engineering model has been constructed with COTS parts. All systems are designed based on competition rules and requirements. Through testing, an asynchronous problem was observed between payload cameras, leading to the failure of the debris detection and data acquisition phase. The suspected cause for this problem is camera adapter software issue. Currently, the team is working on resolving the issue through examination and remediation of software, while also considering alternative adapter options. The testing on debris analysis yielded promising results, indicating that the designed system is feasible. However, measurement error could be further reduced. For future developments, refinements that can be carried out include:

- Resolving camera asynchronous issue;
- Improving measurement accuracy;
- Debris matching through feature matching techniques;
- Developing camera calibration software;
- Identifying more debris features, such as average diameter, colour, shape and albedo.

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